FROST: Frozen Representations for Ocular Screening and Triage in Referable Diabetic Retinopathy

Ritu Chauhan[1]

Abstract

Background: Diabetic Retinopathy (DR) is one of the most prominent causes of preventable vision loss, and the main motive of fundus screening is to determine whether an image is referable and needs specialist review. It is still unclear how various foundation models pretrained differently compare under a matched and statistically rigorous protocol.

Method: seven frozen models were used as backbones which comprised of two supervised (ReseNet-50, ConvNeXt-Base), three general self-supervised learning (Dinov2-Base, DINOv2-Large, DINOv3-Large), and one retina specific SSL (RETFound-Green), all at matched and native resolution. Frozen embeddings were extracted from the images given in BRSET with a patient-level 60/15/15/10 split (test: 1,623 images, 854 patients) for every single backbone separately and two heads were trained per backbone: a shared multi-task head (MT) and a linear probe (LP). Referable DR (grade ≥ 2) was the main focus with a six-task multi-condition macro-AUROC as secondary. Patient-clustered bootstrap 95% confidence intervals (2,000 resamples) have also been reported for the AUROC's and AUPRC's.

Results: even though RETFound-Green has 14 times less parameters than DINOv2-Large (approximately 304M) having only about 22M parameters, RETFound-Green native-392 MT achieved the strongest referable-DR AUROC of 0.985 with 95% CI 0.964-0.997. Hence the smaller retina-trained model was not separable from DINOv2-Large under a paired bootstrap and had only a minor edge over DINOv2-Base (AUROC Δ +0.012, 95% CI 0.004–0.022). The MT's for every frozen model achieved a referable-DR AUROC above 0.97. Scaling within the DINO family from Base to Large or from DINOv2 to DINOv3 did not produce any major noticeable advantage considering frozen extraction. The choice of protocol between native and matched produced a change of +0.032 in macro-AUROC for RETFound which is comparable to changing backbone, making choice of protocol an important performance lever.

Conclusion: Frozen models and the features extracted by then can be used for strong and reproducible referable-DR screening baseline on BRSET. Representation quality without changing the backbone for reusability can matter as much as scale and complexity of the backbone itself. Rankings also take into account the targeted results, head, preprocessing protocols of the models and statistical uncertainty making them more robust and comparative.

Keywords

Diabetic Retinopathy screening, Retinal foundation models, frozen feature extraction, referable DR, BRSET, multi-task learning

1. Introduction

Diabetic retinopathy, a leading cause of preventable vision loss, has the clinical advantage of utilizing a binary triage decision instead of detailed grading to define the significance of fundus screening. During screening, every patient is classified as having referable disease (an individual requiring referral for an ophthalmologist examination and/or treatment) or non-referable disease (an individual safe for continued routine care). The line between the two, typically established at moderate retinopathy or worse, marks the operating point at which sight-threatening retinopathy will be detected prior to visual impairment. Where ophthalmologist scarcity is at its most pronounced, particularly in low- and middle-income countries, the prevalence of diabetes will far exceed ophthalmological examination capacity. A cost-effective and accurate triage that is easily deployable by a technician is, thus, critical for clinical practice. Since an easily acquired non-invasive fundus image can be acquired in mere seconds and is increasingly obtainable with a portable camera and serves as ideal input to such a triage, automating such readings can make the process much more efficient and consistent.

Beyond revealing intraocular conditions, fundus photographs also offer a unique window into systematic pathologies. The living microvascular networks within the retina represent the body's singular view of its in-vivo microvasculature, and the caliber, tortuosity and neural-layer changes apparent within such images reflect the vascular basis for some underlying systemic diseases. The individual image may thus convey risk signals for metabolic or cardiovascular disease, as well as provide insights into ophthalmological abnormalities.

Our use of such systemic signals are strictly as a measure of screening or risk, rather than diagnosis: we note an image labeled as 'hypertensive' fundus changes as indicating ophthalmological grading, not measurement of systemic blood pressure, while an image marked as a "diabetes" flag reflects a representation derived prediction, not a diagnosis derived from fundus evidence alone. Within this framing, fundus reader(s) act as multi-condition screening triage instruments, and the goal is to learn image representation methods that best address the variety of classification objectives.

One means to learn representation for generalizable tasks is by transforming an image to a pre-learned embedding space for a downstream model to use. Three classes of models vie for use in this role when applied to fundus image: supervised convolutional networks (CNNs), which learned visual hierarchies based on supervised natural-image training datasets, general vision-transformer (VTs) models trained using self-supervised approaches such as the DINOv2 and DINOv3 family [1, 2], and retina-specific VTs learned by using self-supervised learning approaches on large unlabeled datasets such as the RETFound-Green series [3]. Each represent different assumptions regarding the source of transferred knowledge: labeling on natural images, the scaling of general image statistics, or supervised pretraining focused on domain-specific learning. Whether or not these different approaches carry any benefit over a scaled, generic approach when the CNN or VT is held fixed at pretraining time, however, remains to be seen, as fixing the model enables comparison of representation quality across varying models while minimizing differences caused by fine-tuning.

Because previous studies vary significantly on metrics such as the images used for training or inference, image augmentation policies, input resolution, definition of primary endpoints, or whether images are sorted at image or patient level, it remains unclear which underlying representations possess the highest generalization potential. Specifically, many previous analyses utilized image-based metrics, whose values may be falsely elevated by repeated images from individual patients included in both training and test sets. We introduce the first controlled study of fundus learning representing the spectrum of prevalent foundation models across general and retina-specific architectures at equivalent resolutions under a patient-level, frozen-extraction design, leveraging patient-clustered bootstrapped confidence interval analysis. We evaluate referable DR as the primary result and various features that can be used for systemic-risk screening. Our specific contributions consist of a controlled patient-level frozen comparison across representation families, heads and protocols with the evaluation being focused on referable-DR along with patient-clustered bootstrap uncertainty and an analysis of whether representation family, head design or protocol explains performance differences.

2. Literature Review

Medical imaging AI is often used for automated DR detection, but they are reported with high discrimination for referable disease on standard datasets and many reach regulatory deployment [4]. This has created a shift in focus of studies from "can a model detect referable DR at all" toward whether reported performance is obtained under conditions that transfer to practice and include patient-level splits, uncertainty measures and reproducible robustness across various devices and populations. Thus, the DR-screening literature indicates referable DR as a clinically meaningful and a more sort after target as compared to the methodology of how to compare representations themselves.

BRSET is a Brazilian multi-label colour-fundus cohort that pairs images with both ophthalmological gradings and patient-level metadata which is why it is commonly used for multi-condition screening and subgroup-aware evaluations [5]. We have used it to get a combination of a DR severity grade along with several binary ocular labels on the same images. BRSET has heavily imbalanced DR grade distribution which makes retina-derived diagnoses its limitation. These properties make BRSET a strong internal substrate but because of its limitations a raw multi-class DR metric becomes misleading which is why we have adopted a referable-DR basis.

Three major families of models compare today: First-generation retina-focused self-supervised models from the RETFound series, pretrained on unlabelled fundus data and evaluated for transferability to ocular and systemic outcomes[6]. Modern general-purpose self-supervised vision transformers like the DINOv2 and DINOv3 series learn general image representations that are applied increasingly to medical use-cases [1, 2]. Classical supervised models, from ResNet-50 to ConvNeXt (that underlies the original BRSET), serve as canonical reference models [7, 8]. Prior literature shows different strengths for both generalist and domain-specific models in ophthalmic applications, that learning quality is non-monotonic with model scale in medical imaging, and that representations learned under supervision or from natural images need careful adaptation [9].

It is precisely this representation quality that our fixed matrix aims to test. Freezing and linear evaluation of pretrained models is the standard approach for assessing the quality of learned representations, and it offers three main practical benefits over full adaptation: The entire performance advantage is attributable to the pretrained representations, performance is reproducible from shared pre-cached features, and computation is feasible on affordable hardware. The community already views frozen evaluation as a standard approach to learning representations, using public releases of frozen embeddings for fundus and other medical cohorts; indeed, one such public release provides DINOv3 features for BRSET [10]. This body of work primarily focuses on individual approaches, rather than on comprehensive comparison of all prevalent paradigm approaches under a consistent protocol applied to a specific clinically relevant outcome; our work is designed to fill this niche. Our closest precursor using the frozen approach, the work of Restrepo and colleagues on Embedding-Based Representations for BRSET and mBRSET, uses the same BRSET test set with provided pre-extracted DINOv3 (ViT-S/16 and ViT-B/16) embeddings and ConvNeXt embeddings under a shared pipeline [10].

Here, we expand the frozen set by including DINOv3-Large, RETFound-Green, and a fully controllable axis of backbone size and training protocol, including a retina-specific backbone with both matched-protocol and native-protocol acquisitions. Comparing directly on the adaptation side, Li et al.'s comparison of DINOv3 [11], RETFound, and VisionFM [12] explores optimization of adaptation for 2-way and 3-way referral of DR and did not evaluate the performance of all possible combinations of represented data in our framework. We question how representational performance varies in light of fixed choices of backbone technology, learned objectives, training, acquisition protocols, patient clustering and the available choices of supervised heads.

Together, the above three axes (represented data, head type, and acquisition protocol) are missing from the fixed protocol in current comparisons. We choose two head architectures as realistic examples of downstream tasks associated with a frozen reader: the linear probe and the multi-task head. The linear probe is a minimal-sized classifier; its output quantifies purely the linear separability of features. The shared multi-task head may harness other binary classes to indirectly aid optimization for a specific sparse target: such additional information may be most helpful when positive case counts are small . We therefore explicitly explore whether shared task supervision enables positive-targeted rather than only aggregate performance improvement in the sparse-positive regime.

Two different heads allow us to disentangle the best architecture ("what features are easiest to classify?"). We often consider confidence estimates of zero, which, if sufficiently tight, confirm robustness against the noise that is inherent in any moderate-sized dataset and in a sample of positive cases within the set of individuals who should be evaluated on this clinical outcome. For most comparisons to date using clinical data with multiple images per subject, confidence intervals have used the images as the units of resampling and this leads to underestimating uncertainty when multiple images may convey redundant information about a subject; it is critical to center uncertainty estimates around the patient, and this is what we do for all tests.

Table 1. Comparative summary and research gaps in prior retinal foundation-model and diabetic retinopathy screening studies.

Title and Author(s)	Author's work	Gap	Gap filled by my research
BRSET: A Brazilian Multilabel Ophthalmological Dataset of Retina Fundus Photos [5]; Deep Learning for Comprehensive Analysis of Retinal Fundus Images: Detection of Systemic and Ocular Conditions [13]	Evaluated supervised CNNs and transformers on BRSET for DR and selected ocular/systemic-proxy tasks.	No controlled frozen comparison across supervised, general self-supervised, and retina-specific self-supervised representations. The original BRSET study does not explicitly establish a patient-level comparison protocol, and uncertainty reporting is limited.	A frozen backbone × head matrix evaluated on a patient-level split, comparing supervised CNNs, general SSL transformers, and retina-specific SSL models with patient-clustered bootstrap confidence intervals.
A Foundation Model for Generalizable Disease Detection from Retinal Images[6]; Training a High-Performance Retinal Foundation Model with Half-the-Data and 400 Times Less Compute [3]	Introduced retina-specific foundation models and evaluated transfer through downstream adaptation.	Frozen representation behaviour and a controlled matched-protocol comparison against general-purpose vision models are under-reported.	RETFound-Green is evaluated as a frozen feature extractor at both matched and native resolution, directly alongside general SSL and supervised backbones.
Embedding-Based Representations for BRSET and mBRSET [10]	Released standardized inference-only embeddings using DINOv3 ViT-S/16, ViT-B/16, ConvNeXt-Tiny, and ConvNeXt-Base.	No DINOv3-Large, retina-specific SSL model, prediction-head comparison, or acquisition-protocol comparison.	Adds DINOv3-Large, RETFound-Green, ResNet-50, ConvNeXt-Base, multi-task versus linear-probe heads, and matched versus native RETFound protocols within one patient-level evaluation.
Comparison of Foundation Models and Transfer Learning Strategies for Diabetic Retinopathy Classification [11]	Compared DINOv3, RETFound, and VisionFM for binary and three-class DR using different adaptation strategies.	Frozen representation quality is not isolated within a fixed backbone × head × protocol matrix.	Uses referable DR as the primary endpoint and compares frozen representations using identical heads, patient-level splitting, and paired patient-clustered confidence intervals.
DINOv2: Learning Robust Visual Features without Supervision [1]; DINOv3 [2]	Developed DINOv2 and DINOv3 as transferable general self-supervised visual representations.	DINOv3-Large has not been established on BRSET under this specific frozen multi-task fundus protocol.	Evaluates DINOv3-Large under the same frozen extraction procedure, patient split, prediction heads, and endpoints as every other backbone.
Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs [4]; Comparison of Foundation Models and Transfer Learning Strategies for Diabetic Retinopathy Classification [11]	Demonstrated high referable-DR discrimination using end-to-end or adapted deep-learning models.	Frozen-feature referable-DR evaluation with patient-clustered uncertainty estimates for every backbone–head combination remains scarce.	Reports referable-DR AUROC and AUPRC with patient-clustered bootstrap confidence intervals for every experimental cell.
Synergic Adversarial Label Learning for Grading Retinal Diseases via Knowledge Distillation and Multi-task Learning [14]; Multi-task Deep Learning for Glaucoma Detection from Color Fundus Images [15]	Used shared representations across related retinal tasks to support disease prediction.	The effect of a multi-task head is rarely contrasted directly with a linear probe using exactly the same frozen image embeddings.	Directly compares a shared multi-task head with a per-task linear probe on identical frozen embeddings, allowing endpoint-specific analysis of non-linear and multi-task benefit.

Embedding-Based Representations for BRSET and mBRSET [10]; broader retinal preprocessing/protocol-comparison literature	Standardized preprocessing or compared selected preprocessing choices across retinal image models.	Protocol is rarely measured as a formal experimental axis in retinal foundation-model comparisons.	Matched-224 and native-392 RETFound rows quantify protocol sensitivity; the observed +0.032 six-task macro-AUROC shift is reported descriptively pending expanded confidence-interval coverage.
RetBench: Which Ophthalmic Foundation Model Performs Best and Why? [9]	Benchmarked multiple ophthalmic foundation models across retinal and systemic prediction tasks.	Backbone rankings can still be interpreted from point estimates even when apparent differences may fall within sampling variability.	Uses paired bootstrap testing and explicitly reports when two backbones are statistically "not separable," rather than presenting every numerical difference as a true ranking.
Comparison of Foundation Models and Transfer Learning Strategies for Diabetic Retinopathy Classification, and related cross-device/cross-population DR-transfer studies [11]	Evaluated transfer between retinal datasets, populations, and camera domains.	External validation is not included in the present experimental scope.	Restricts claims to internal BRSET performance and defers cross-population and cross-device transfer claims to future work.

We have targeted a controlled, frozen comparison that tests representation family, head architecture and acquisition protocol on a patient level split and evaluated on a referable DR as the primary endpoint with bootstrap intervals which fills a very narrow and specific gap. This design also allows a detailed assessment of the methodology, scale, head and protocol against the uncertainty that determines whether an apparent difference is real.

3. Methodology

The frozen retinal representations were evaluated in a controlled matrix containing both the frozen backbone and the head, holding the data split, the training regime and random seed fixed while varying only the backbone/protocol that produced the embedding and/or the prediction head trained on top of it. Keeping the pretrained weights of the backbones fixed ensured that any performance difference identified would be due to pretrained representation rather than being attributed to fine-tuning dynamics. This made the comparison exactly reproducible from cached features. The matrix comprised seven backbone/protocol rows and two heads each evaluated on the same held-out test set. Figure 1 shows that the data split, training regime and random seed are held fixed while only the frozen backbone and the prediction heads are changed and the embeddings are only extracted once per backbone and are cached for the two heads.

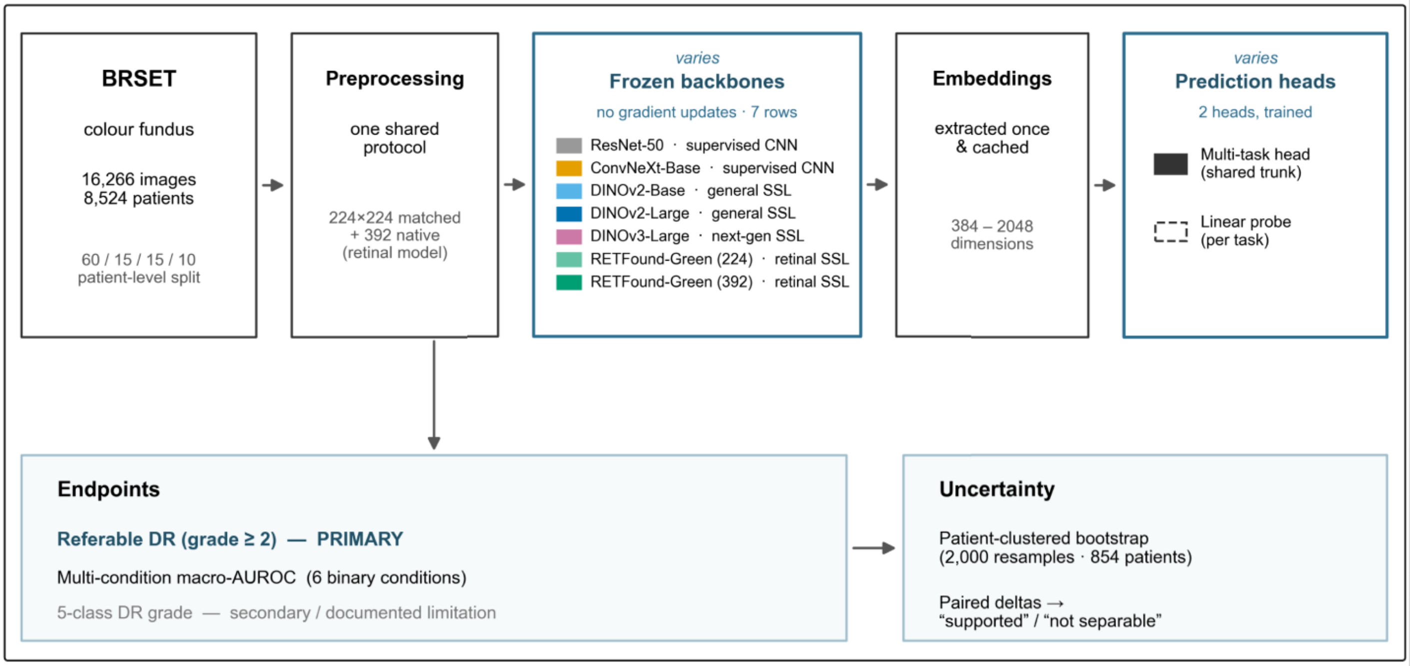

Figure 1. Study design overview

3.1 About the dataset

This study makes use of BRSET (Brazilian Multilabel Ophthalmological Dataset) which contains Brazilian color-fundus cohort with paired ophthalmological gradings and patient-level metadata. This dataset addresses the scarcity of publicly available ophthalmological datasets in Latin America. BRSET comprises 16,266 color fundus retinal photos from 8,524 Brazilian patients, aiming to enhance data representativeness, serving as a research and teaching tool. It contains sociodemographic information, enabling investigations into differential model performance across demographic groups.

Data from three São Paulo outpatient centers yielded demographic and medical information from electronic records, including nationality, age, sex, clinical history, insulin use, and duration of diabetes diagnosis. A retinal specialist labeled images for anatomical features (optic disc, blood vessels, macula), quality control (focus, illumination, image field, artifacts), and pathologies (e.g., diabetic retinopathy).

BRSET comprises 65.1% Canon CR2 and 34.9% Nikon NF5050 images. 61.8% of the patients are female, and the average age is 57.6 years. Diabetic retinopathy affected 15.8% of patients, across a spectrum of disease severity.

A patient-level split was utilized because the cohort contains multiple images per patient in order to prevent leakage of a patient’s images across partitions, using a 60/15/15/10 division for training, validation, reliability and test partitions (reliability partition was reserved for subgroup reliability work outside the scope of this paper), thus the resulting partitions contain 9,763, 2,443, 2,437, and 1,623 images respectively. All model selection used the validation partition only and every metric was computed once on the untouched test partition of 1,623 images from 854 patients.

Table 2. A summary of the dataset.

Property	Value
Source	BRSET; color fundus photography
Total images / patients	16,266 / 8,524
Split scheme	Patient-level 60/15/15/10 (train/validation/reliability/test)

Train	9,763 images / 5,114 patients
Validation	2,443 images / 1,278 patients
Reliability (held out, not used here)	2,437 images / 1,278 patients
Test	1,623 images / 854 patients
Patient overlap across splits	0
Label set	5-class DR grade; six binary conditions (macular edema, hypertensive retinopathy, AMD, drusen, other ocular, diabetes)

3.2 Task and endpoint definitions

The primary result is referable diabetic retinopathy which can be used for binary distinction between referable eyes (grade ≥ 2) and non-referable eyes, this distinction helps in determining whether a screening requires further professional review or not. The referable DR score is produced with the help of the DR grade output of the model that consists of 5 classes, and the score is the total predicted probability mass on classes 2,3 and 4. Because of its severe class imbalance, the 5-class DR grade has been considered a secondary result. There are six binary conditions and they have been proposed in the form of a multi-condition panel and among those, diabetes label is not a direct retina derived diagnosis, and hypertensive retinopathy is not a measurement of systemic blood pressure but an ophthalmological retinal grading.

3.3 Image preprocessing and feature extraction

To make embeddings comparable across backbones, all images were processed under a single resolution-matched protocol at 224 × 224 pixels before extraction. Because the retina-specific backbone was pretrained at its own native resolution, we additionally extracted it under a native protocol at 392 × 392 pixels with average pooling over patch tokens, giving a within-backbone comparison of matched versus native-resolution features. Under the matched protocol, transformer backbones used their class-token embedding and convolutional backbones used global average pooling; positional embeddings for the native-392 weights were interpolated to 224 under the matched protocol. Embeddings were extracted once per protocol and cached, so that all downstream head training and evaluation read fixed feature vectors. Exact resize interpolation, crop policy, pixel scaling, normalization, and deterministic extraction settings are reported in the Supplementary Methods.

All images were processed under a single resolution matched protocol at 224 × 224 pixels before extraction so that the embeddings are comparable across all backbones irrespective of their pretraining protocol.

3.4 Backbone models and prediction heads

The seven backbone rows consist of models from four different pretraining paradigms. Of those, ResNet-50(2048-dimensional features) and ConvNeXt-Base (1024 dimensional) are supervised convolutional networks; DINOv2-Base (ViT-Base, 768-dimensional), DINOv2-Large (ViT-Large, 1024-dimensional), and the next-generation DINOv3-Large (ViT-Large, 1024-dimensional) are self-supervised learning (SSL) vision transformers and the remaining rows are a retina specific SSL model, RETFound-Green (a ViT-Small/14 backbone with 384-dimensional features), evaluated under both the matched (224) and native (392) protocols. BRSET was not present in RETFound-Green's pretraining datasets.

Two types of heads are created over the frozen backbone for the predictions, the linear probe and the multi-task head. The linear head is a simple single linear layer that is trained per task directly on the frozen embeddings produced by the backbone and its main purpose is to measure the raw linear separability of the representation. The multi-task head is more complex than the linear head and has a single common trunk with per task output layers trained jointly and with per-task losses that are combined together using kendall uncertainty weighting so that the model learns the relative weight of each task instead of just using fixed weights. Different losses were used according to their tasks like cross entropy for the 5-class DR grade and binary cross-entropy for the six binary conditions.

3.5 Training procedure

A single pre-specified recipe was applied identically to all fourteen cells, so that any performance difference reflects the backbone or head rather than tuning. Heads were trained with AdamW at a learning rate of 1×10^{-4}, weight decay of 0.01, batch size 256, and cosine annealing after a five-epoch linear warmup, with gradient clipping at 1.0 and a cap of 100 epochs. Model selection used early stopping on validation multi-condition macro-AUROC with patience 10, and the best-validation checkpoint was restored before test evaluation. Validation macro-AUROC was pre-specified for checkpoint selection to avoid selecting epochs on the rare primary endpoint alone and to preserve a common criterion across the multi-condition panel. No class weighting was applied; all runs used a fixed seed of 42. Imbalance mitigation was deliberately excluded from this comparison and reserved for a separate ablation, so that the matrix measures representations under one neutral recipe.

3.6 Evaluation and statistical analysis

For each cell we have computed per-task area under the receiver operating characteristic curve (AUROC), are under the precision-recall curve (AUPRC) for sparse-positives, and multi-condition macro-AUROC as a summary of the binary tasks, the ordinal DR grade was reported separate from the aggregate. All metrics were calculated at the image level and patients were considered as clustered and resampled during uncertainty estimation to ensure that all the images from a particular patient were retained together

A patient-clustered bootstrap was used for calculating uncertainty where 854 test patienst were resampled with replacement and each metric was calculated again for 2,000 resamples and the 2.5th and 97.5th percentiles were taken as the 95% confidence interval (CI) (all done on a fixed seed) further visualized in figure 2.

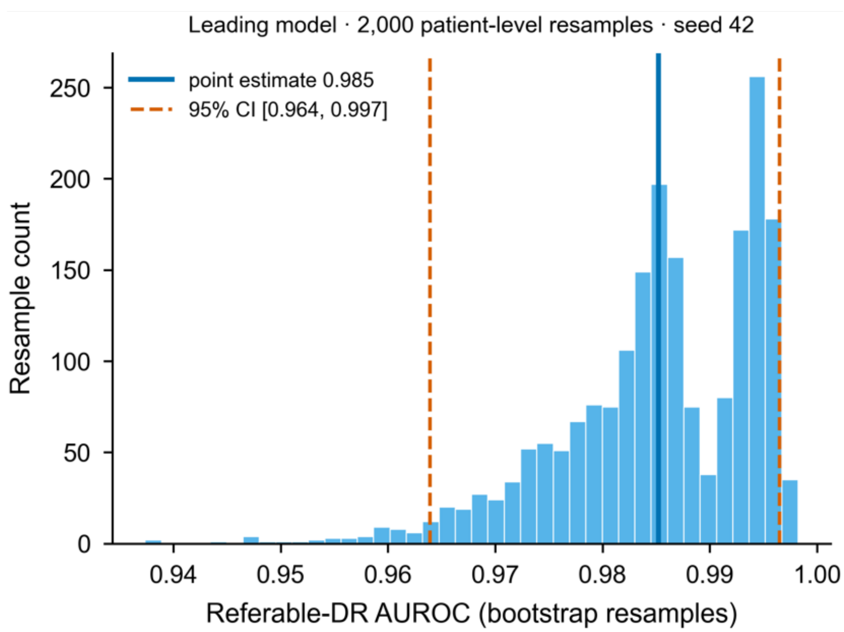

Figure 2. Patient clustered bootstrap distribution of referable-DR AUROC for the leading model (RETFound-green)

4. Results

4.1 Test cohort and label distribution

Out of 1,623 test images, referable DR is present in 73 (4.50%), and the underlying DR grade is dominated by grade 0 (1,517 images, 93.5%), with mild, moderate, severe, and proliferative grades at 33, 11, 22, and 40 images respectively as shown in Figure 3 and Table 3. The binary panel ranges from densely to very sparsely positive: drusen (261 positives, 16.1%), diabetes (230, 14.2%), and other ocular (143, 8.8%) are well populated, whereas macular edema (33, 2.0%), hypertensive retinopathy (30, 1.8%), and AMD (22, 1.4%) are sparse. These sparse endpoints yielded wider patient-clustered bootstrap confidence intervals and therefore warrant precision–recall reporting and cautious cross-model interpretation. The extreme DR-grade imbalance is precisely why we adopt referable DR rather than fine-grained severity as the primary endpoint.

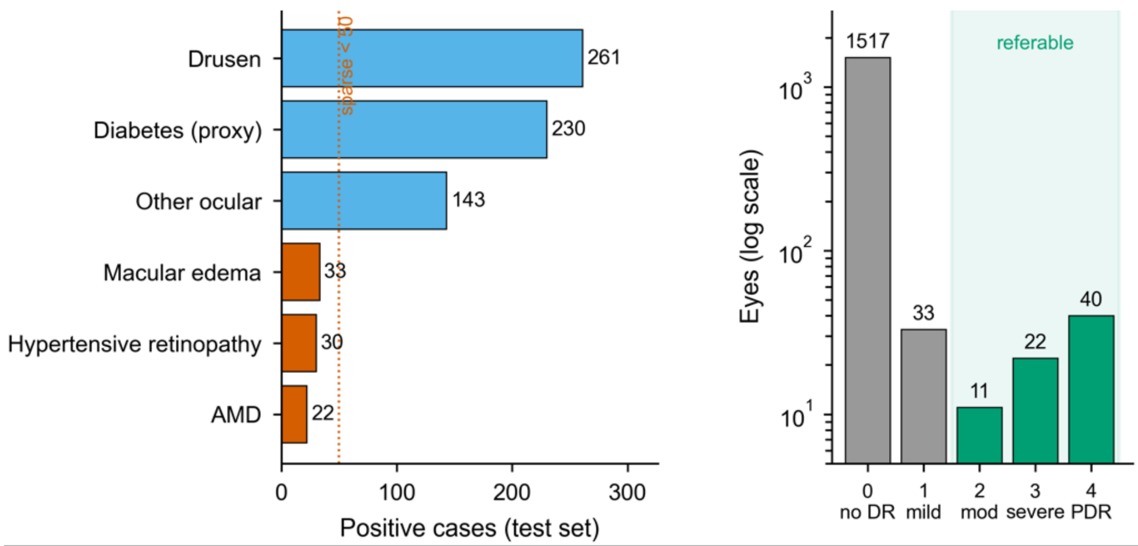

Figure 3. Cohort and label distribution of BRSET

Table 3. cohort and task characteristics (test partition, n = 1,623 images / 854 patients)

Task	Positives	Prevalence	Notes
Referable DR	73	4.50%	Primary result; from DR-grade probability mass on grades 2-4
DR grade (5-class)	grade 0/1/2/3/4 = 1,517 / 33 / 11 / 22 / 40	93.5% grade 0	Secondary; severe imbalance
Drusen	261	16.1%	Well populated
Diabetes	230	14.2%	Record-derived proxy label
Other Ocular	143	8.8%	Heterogeneous composite
Macular edema	33	2.0%	Sparse
Hypertensive retinopathy	30	1.8%	Sparse; retinal grading, not systemic measurement
AMD	22	1.4%	Sparsest; wider bootstrap CIs

4.2 Primary referable-DR result

Six of the seven total multi-task cells achieved referable-DR AUROC above 0.97 where ResNet-50 performed worst at 0.942 (95% CI 0.901-0.974) and RETFound-Green (392, native) performed best getting an AUROC of 0.985 (95% CI 0.964-0.997). DINOv2-Large and DINOv3-Large also performed well even after not being specific to retinal imaging based pretraining getting AUROC's of 0.979 (0.956-0.993) and 0.977 (0.956-0.993) respectively.

RETFound-Green native-392, DINOv2-Large, and DINOv3-Large were not separable under paired bootstrap unlike DINOv2-Base that was noticeably weaker than RETFound-Green native-392. High performance was also seen in precision and recall metrics but with more separation as shown in Figure 5a, and the strongest cell reaching referable-DR AUPRC of 0.882 (0.797-0.941) against a 4.5% positive base rate as shown in figure 5b.

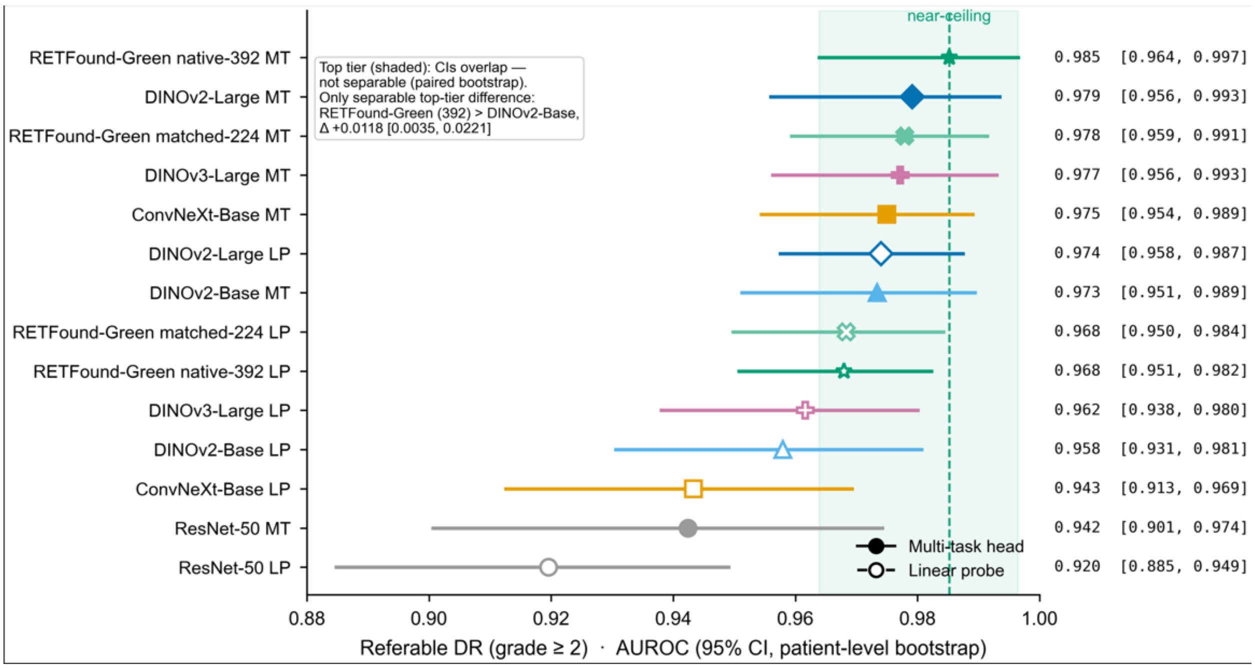

Figure 4. Primary referable-DR performance across the matrix

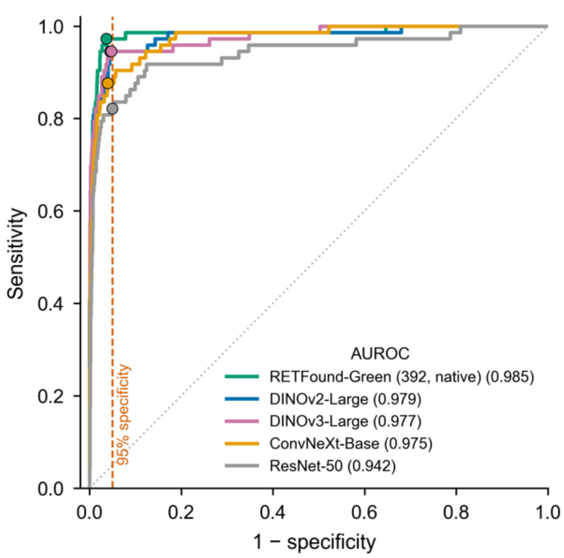

Figure 5a. Per sample receiver operating characteristic curve; dots representing sensitivity at 95 % specificity

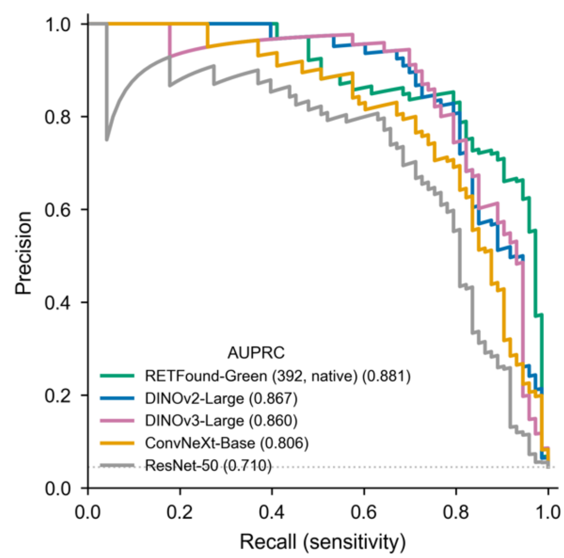

Figure 5b. Precision-recall curves against the 4.5 % positive base rate

4.3 Parameter-efficiency and the domain-pretraining frontier

Despite being the smaller retina-specific model, RETFound-Green (392, native), a ViT-Small backbone with 384-dimensional embeddings got the only referable-DR backbone difference over DINOv2-Base with the multi-task head in the bootstrap (AUROC Δ +0.012; 95% CI 0.004–0.022). Even under its native extraction protocol it achieved the same performance as the much larger general ViT-Large models like DINOv2-Large (0.006 Δ) and DINOv3-Large (0.008 Δ), both not separable under paired bootstrap as shown in Table 5.

The native RETFound comparison incorporates differences in pretraining domain, input resolution, pooling strategy, and positional-embedding handling in addition to model scale. It therefore demonstrates the performance of the retina-specific model under its intended native protocol rather than isolating a pure domain-pretraining-versus-scale effect. Nevertheless, RETFound-Green uses approximately 22M parameters, compared with approximately 304M for DINOv2-Large, and produces 384-dimensional rather than 1024-dimensional embeddings, yielding a smaller feature cache and lighter downstream head (Figure 6). We did not run a formal non-inferiority test, so we report the supported edge over DINOv2-Base and non-separation from the larger models without claiming equivalence.

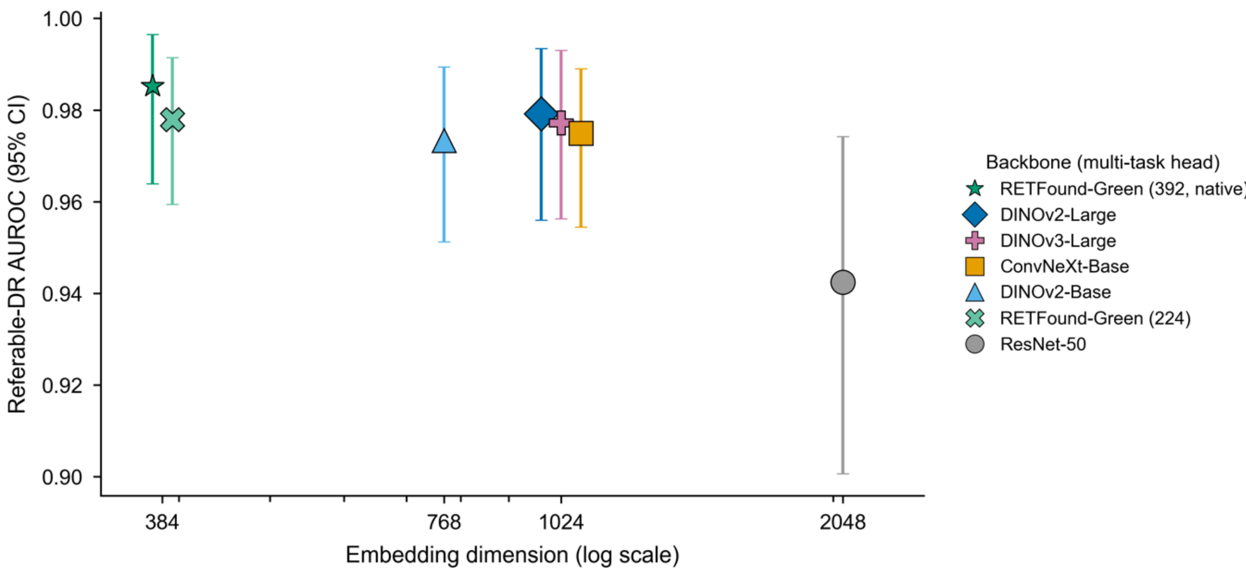

Figure 6. Referable-DR AUROC against frozen-feature embedding dimensions; points sharing a dimension are offset horizontally for legibility

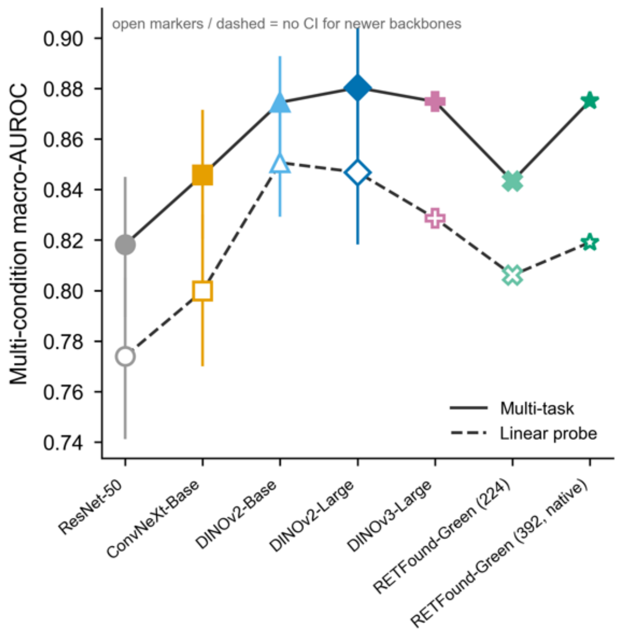

Figure 7a. Macri-AUROC (mean AUROC over six binary conditions) for each backbone, multi-task (solid) and linear-probe (dashed).

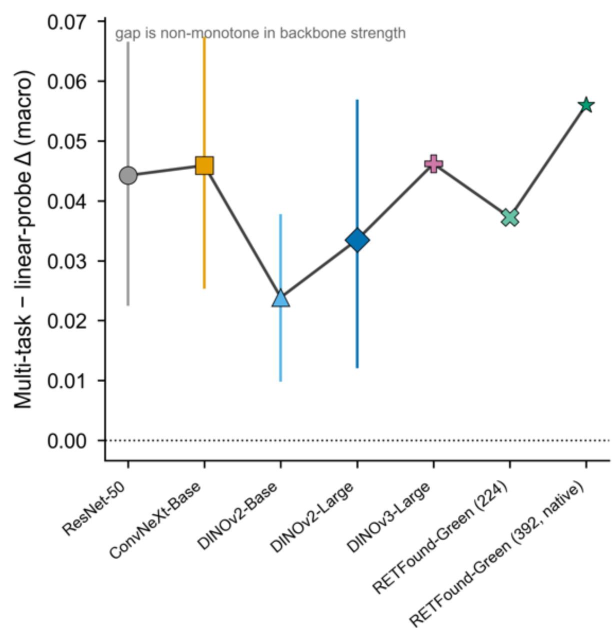

Figure 7b. The multi-task minus linear-probe macro gap is not monotone in backbone strength; DINOv2-Base shows the smallest gap.

4.4 Backbone axis: paradigm over scale, and recency does not dominate

We have identified that the pretraining paradigm matters more than the scale of the models that have been used in case of multi-condition macro-AUROC where confidence intervals are available for the four original backbones (see Figure 8). This can be seen when moving from the supervised convolutional ConvNeXt-Base to self-supervised DINOv2-Base raised the macro-AUROC by +0.029 (Figure 7a), whereas scaling within the self-supervised family from DINOv2-Base to DINOv2-Large added only +0.006 which was not separable under a paired bootstrap as shown in Table 5. DINOv3-Large did not show a separable jump in referable DR as compared to DINOv2-Large (-0.002), hence proving that model recency also did not dominate.

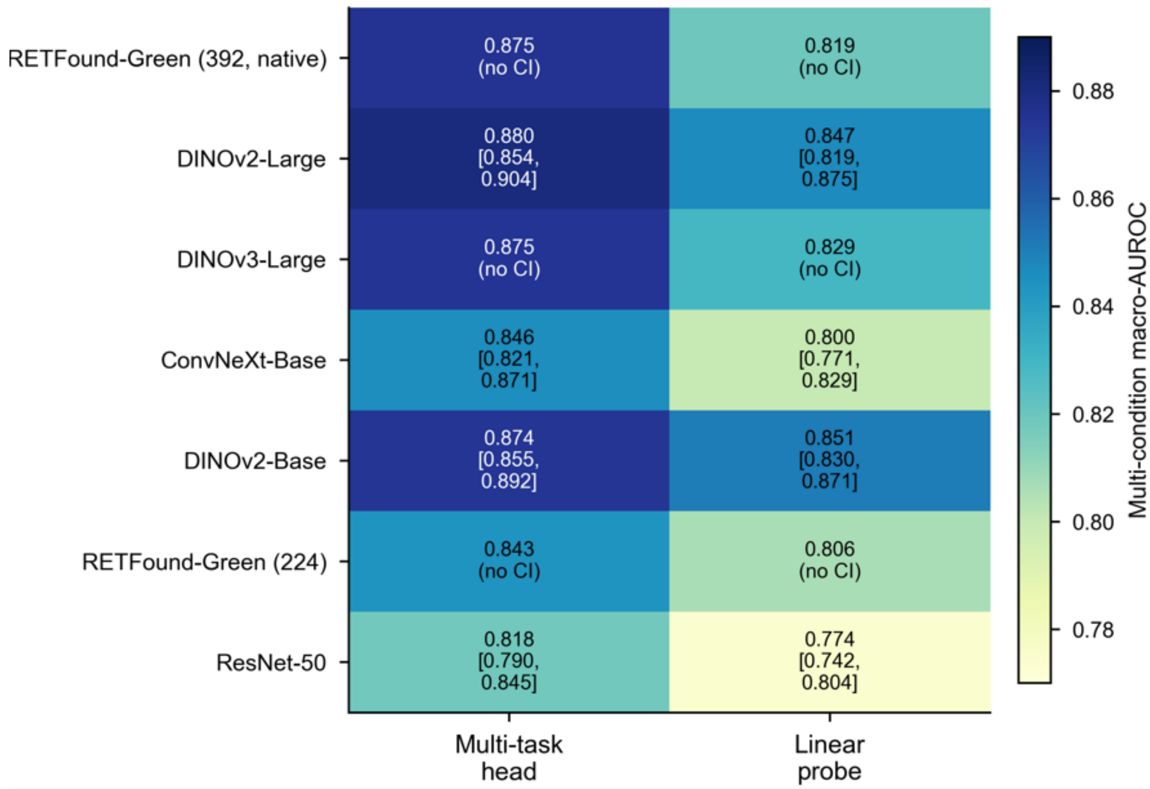

Figure 8. Multi-condition macro-AUROC for every backbone and head type

4.5 Multi-condition screening and the head axis

The multi-task head showed improvement over the linear probe heads especially on the sparse conditions. The multi-task head performed better than the linear probes for every backbone on multi-condition macro-AUROC like ResNet-50 +0.044 (0.023–0.066), ConvNeXt-Base +0.046 (0.026–0.067), DINOv2-Base +0.024 (0.010–0.038), and DINOv2-Large +0.033 (0.012–0.057). The difference between the performance of the two heads was largest on AMD which is the rarest condition and the difference reached +0.131 (0.057–0.215) for ResNet-50 and +0.109 (0.015–0.212) for DINOv2-Large as shown in Table 5.

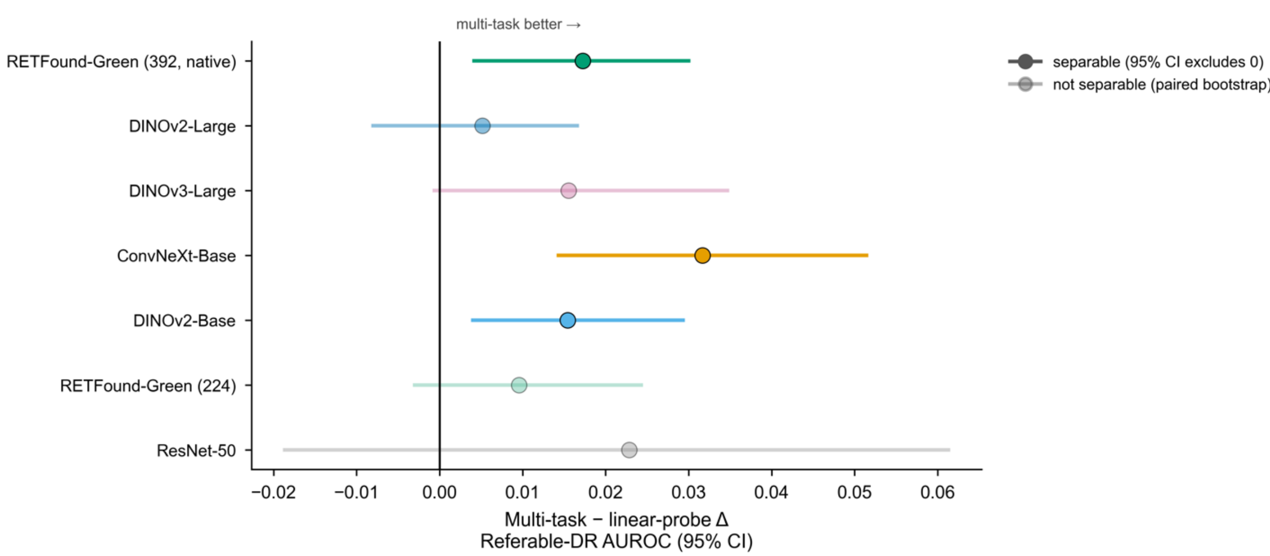

Figure 9. Per-backbone difference in referable-DR AUROC between the multi-task head and the linear probe

The improvement was not caused by just the backbone strength since DINOv2-Base showed the narrowest head gap and DINOv2-Large a wider one (Figure 7b), so stronger backbones did not steadily reduce the value of the multi-task head. On referable DR, the effect caused by the heads was dependent on the result and was not separable for cells already at 0.97+ (Figure 9). Although the multi-task head's advantage was supported for all seven backbones when calculating precision-recall.

4.6 Protocol sensitivity: a within-backbone change rivals a change of backbone

Changing the acquisition protocol created a difference in the retina-specific model by an amount equivalent to a change of backbone. Multi-condition macro-AUROC increased by 0.032 by switching RETFound-Green from the matched-224 to the native-392 protocol and this shit is comparable to the change from CNN to transformer protocol (+0/029) and is also larger than the within family change (+0.006) (Figure 10). Referable-DR AUROC increased by 0.007 which was not separable under bootstrap by its AUPRC gain was 0.048 which was supported (Figure 11; Table 5). Hence the native protocol performed the strongest out of all the cells and had a supported precision-recall improvement thus showing that preprocessing and feature access choices are not fixed details but can be experimented with.

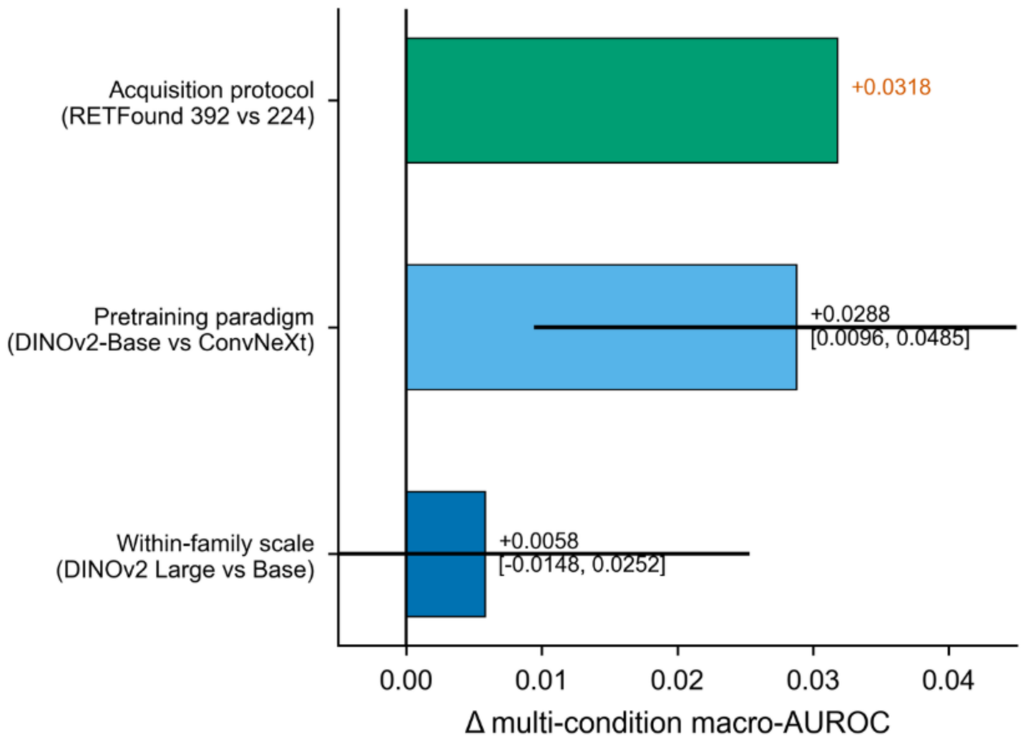

Figure 10. Magnitude of backbone-attributable effects on multi-condition macro-AUROC. Changing the retina-specific model's input protocol (native-392 vs matched-224) shifts macro-AUROC by more than the pretraining-paradigm or within-family scale contrasts.

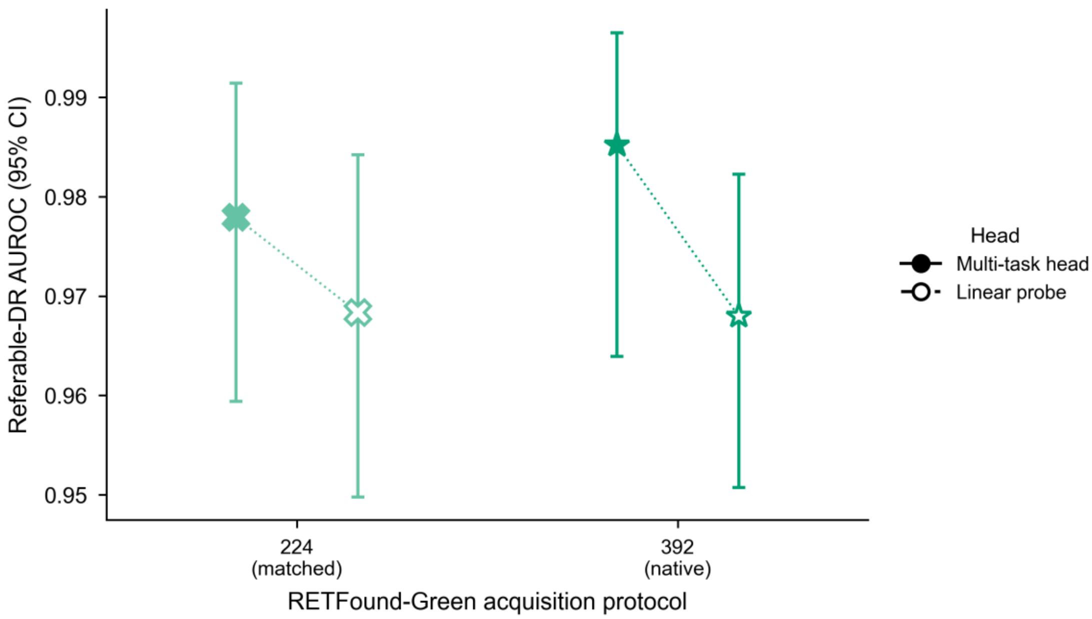

Figure 11. Referable-DR AUROC for RETFound-Green at matched (224) and native (392) protocols; The two heads are offset horizontally at each protocol for legibility. The native protocol yields the strongest single cell but is not separable from matched extraction on referable-DR AUROC under paired bootstrap.

4.7 DR severity (5-class): a strong screening endpoint with weak fine-grained grading

Fine-grained DR grading was the matrix's weakest result caused due to the class imbalance in DR grade in BRSET. Grade 0 got 93.5% correct predictions while mild and moderate grades received essentially no correct predictions so the overall balanced accuracy stayed well below what was expected ranging from 0.384 for ResNet-50 multi-task head to 0.474 for DINOv2-Large multi-task head.

Table 4. Master matrix

Backbone (protocol)	Head	Referable-DR AUROC [95 % CI]	Multi-condition macro-AUROC	5-class bal. Acc.
ResNet-50 (224)	MT	0.942 [0.901–0.974]	0.818 [0.790–0.845]	0.384
ResNet-50 (224)	LP	0.920 [0.885–0.949]	0.774 [0.742–0.804]	0.239
ConvNeXt-Base (224)	MT	0.975 [0.954–0.989]	0.846 [0.821–0.871]	0.400
ConvNeXt-Base (224)	LP	0.943 [0.913–0.969]	0.800 [0.771–0.829]	0.307
DINOv2-Base (224)	MT	0.973 [0.951–0.989]	0.874 [0.855–0.892]	0.436
DINOv2-Base (224)	LP	0.958 [0.931–0.981]	0.851 [0.830–0.871]	0.418
DINOv2-Large (224)	MT	0.979 [0.956–0.993]	0.880 [0.854–0.904]	0.474
DINOv2-Large (224)	LP	0.974 [0.958–0.987]	0.847 [0.819–0.875]	0.423
RETFound-Green (224, matched)	MT	0.978 [0.959–0.991]	0.843 (PE)	0.399
RETFound-Green (224, matched)	LP	0.968 [0.950–0.984]	0.806 (PE)	0.355
DINOv3-Large (224)	MT	0.977 [0.956–0.993]	0.875 (PE)	0.419
DINOv3-Large (224)	LP	0.962 [0.938–0.980]	0.829 (PE)	0.347
RETFound-Green (392, native)	MT	0.985 [0.964–0.997]	0.875 (PE)	0.463

RETFound-Green (392, native)	LP	0.968 [0.951–0.982]	0.819 (PE)	0.332

Table 5. Paired deltas with verdicts

Comparison	Metric	Δ [95% CI]	Bootstrap verdict
Backbone (multi-task)			
ConvNeXt-Base → DINOv2-Base	Macro-AUROC	+0.029 [0.010–0.049]	supported
DINOv2-Base → DINOv2-Large	Macro-AUROC	+0.006 [−0.015–0.025]	not separable
RETFound-Green (392) vs DINOv2-Base	Referable-DR AUROC	+0.012 [0.004–0.022]	supported
RETFound-Green (392) vs DINOv2-Large	Referable-DR AUROC	+0.006 [−0.017–0.031]	not separable
RETFound-Green (392) vs DINOv3-Large	Referable-DR AUROC	+0.008 [−0.003–0.022]	not separable
DINOv3-Large vs DINOv2-Large	Referable-DR AUROC	−0.002 [−0.021–0.018]	not separable
Head (multi-task − linear probe)			
ResNet-50	Macro-AUROC	+0.044 [0.023–0.066]	supported
ConvNeXt-Base	Macro-AUROC	+0.046 [0.026–0.067]	supported
DINOv2-Base	Macro-AUROC	+0.024 [0.010–0.038]	supported
DINOv2-Large	Macro-AUROC	+0.033 [0.012–0.057]	supported
ResNet-50	AMD AUROC	+0.131 [0.057–0.215]	supported
DINOv2-Large	AMD AUROC	+0.109 [0.015–0.212]	supported
ConvNeXt-Base	Referable-DR AUROC	+0.032 [0.014–0.051]	supported
DINOv2-Base	Referable-DR AUROC	+0.015 [0.004–0.029]	supported
DINOv2-Large	Referable-DR AUROC	+0.005 [−0.008–0.017]	not separable
Protocol (RETFound-Green, native − matched)			
Native-392 vs matched-224 (MT)	Referable-DR AUROC	+0.007 [−0.004–0.020]	not separable
Native-392 vs matched-224 (MT)	Referable-DR AUPRC	+0.048 [0.003–0.096]	supported

5. Discussion and limitations of the study

Strong Reproducible referable-DR screening can be done on BRSET using frozen retinal foundation-model features without changing pretrained weights. A lightweight head trained on embeddings produced by frozen models without any update to the backbone is able to differentiate between referable and non-referable eyes at up to 0.985 (95 % CI 0.964-0.997) and all rows exceeding 0.97. Hence practically the backbone which is the most expensive part to train both computationally and in terms of time taken can be used as is with an additional head that is cheap to fit and modify and is reproducible because the features are extracted by the fixed backbone once and then stored and reused.

This is very important clinically because this shows that a reusable frozen representation along with a lightweight retrainable triage head can be used to prioritize specialist review for fundus images in resource-constrained and scarce ophthalmologist conditions

The retina-specific model (under its native extraction protocol) performed similar to the general ViT-Large models even after being much smaller in terms of trainable parameters. The ViT-Small retina specific model has the backbone advantage for referable-DR and a slight edge over DINOv2-Base but is not separable from ViT-Large general models above it. This result matches the expectations considering domain-specific pretraining under frozen extraction. This learning can be applied practically to reduce feature-cache storage and creating lighter downstream heads where the retina-specific model can determine predictions of the target result requiring much smaller cached embeddings and lower-cost head retraining.

This result shows that neither the scale of the trainable parameters nor the model recency alone is a safe factor for determination of a default when selecting a frozen backbone for fundus screening. A much newer next-generation general purpose transformer performed almost equivalent on referable DR as compared to its previous versions. Hence, backbone selection must be done empirically based on the target endpoint and not on the basis of the model's scale or release date.

The head axis is best read as a task-structure decision rather than a generic upgrade. The multi-task head showed targeted gains over linear probing that concentrated on sparse-positive conditions, most visibly AMD. Separately, the magnitude of the multi-task-versus-linear-probe gap was not monotone in backbone strength, indicating that stronger frozen backbones did not steadily reduce the value of the more expressive multi-task head. On referable DR, an endpoint already near the discrimination ceiling, the multi-task head added little additional ranking accuracy but still improved precision–recall. The multi-task head is therefore particularly useful where sparse targets may benefit from a shared representation, while it is a more marginal choice on an endpoint where discrimination is already saturated.

Acquisition or extraction protocol (resolution, pooling, and positional-embedding handling between matched-224 and native-392 extraction) also created a shift in the multi-condition macro by an amount almost equivalent to that created by the change of backbone and also produced a supported precision-recall gain on referable-DR. This makes it both a methodological contribution and an experimental result and preprocessing must be considered during model selection.

The endpoint structure of the study is internally consistent and disciplines what should be headlined. Strong referable-DR discrimination coexisted with weak five-class grading from the same frozen features, which is expected: separating referable from non-referable eyes is a genuinely easier problem than ranking all five ordinal grades on a distribution dominated by a single class. This coexistence justifies referable DR as the primary DR endpoint and argues against headlining multi-class accuracy on a heavily imbalanced grade distribution, where a high aggregate number can hide collapse on the rare, clinically important grades.

This study also contributes to paired uncertainty that affects how the comparisons provided should be read. When computed at the patient level, the differences between the backbones ended up being within sampling noise and were reported as "not separable". This study adds a much broader backbone and pretraining coverage over the previous work including the frozen BRSET embedding release [10] and the adaptation-focused BRSET comparison [11] also adding an explicit head axis, an extraction-protocol axis and patient-clustered paired testing.

6. Conclusion

We reported a controlled, patient-level, frozen backbone × head evaluation on BRSET, with referable DR (grade ≥ 2) as the primary endpoint and patient-clustered bootstrap confidence intervals for every cell. Frozen foundation-model features provide a strong, reproducible referable-DR screening baseline that a lightweight head can realise without fine-tuning the backbone.

The instructive result is not a single universal winner: the leading models were largely not separable under paired bootstrap, while the only supported top-tier advantage was held by the smaller retina-specific model. Under its native extraction protocol, that model reached the same referable-DR performance regime as much larger general transformers. Taken together, the findings show that representation quality, accessed efficiently through frozen embeddings, can matter as much as parameter scale, model recency, or head complexity, while extraction protocol can rival architecture choice as a lever on measured performance.

A frozen fundus reader's ranking is therefore not a fixed property of its backbone but a joint function of the endpoint, downstream head, extraction protocol, and statistical uncertainty against which it is judged. Establishing whether this strong internal screening baseline holds under population and device shift is the next step.

7. References

1. Oquab, M. et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research* (2024).

2. Siméoni, O. et al. DINOv3. *arXiv* preprint arXiv:2508.10104 (2025). https://doi.org/10.48550/arXiv.2508.10104

3. Engelmann, J. & Bernabeu, M. O. Training a high-performance retinal foundation model with half-the-data and 400 times less compute. *Nature Communications* 16, 6862 (2025). https://doi.org/10.1038/s41467-025-62123-z

4. Gulshan, V. et al. Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *JAMA* 316, 2402–2410 (2016). https://doi.org/10.1001/jama.2016.17216

5. Nakayama, L. F. et al. BRSET: A Brazilian multilabel ophthalmological dataset of retina fundus photos. *PLOS Digital Health* 3, e0000454 (2024). https://doi.org/10.1371/journal.pdig.0000454

6. Zhou, Y. et al. A foundation model for generalizable disease detection from retinal images. *Nature* 622, 156–163 (2023). https://doi.org/10.1038/s41586-023-06555-x

7. He, K., Zhang, X., Ren, S. & Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* 770–778 (2016). https://doi.org/10.1109/CVPR.2016.90

8. Liu, Z. et al. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* 11976–11986 (2022). https://doi.org/10.1109/CVPR52688.2022.01167

9. Zou, K. et al. RetBench: Which ophthalmic foundation model performs best and why? In Fang, H. et al. (eds) *Ophthalmic Medical Image Analysis (OMIA 2025)*. *Lecture Notes in Computer Science*, vol. 16209, 75–84 (Springer, 2026). https://doi.org/10.1007/978-3-032-10351-2_8

10. Restrepo, D., Wu, C., Morley, M., Celi, L. A. & Nakayama, L. F. Embedding-Based Representations for BRSET and mBRSET, version 1.0.0. *PhysioNet* (2026). https://doi.org/10.13026/1h4p-vz70

11. Li, L. Y., Lebiecka-Johansen, B., Byberg, S., Thambawita, V. & Hulman, A. Comparison of foundation models and transfer learning strategies for diabetic retinopathy classification. *medRxiv* (2026). https://doi.org/10.64898/2026.04.17.26351092

12. Qiu, J. et al. Development and validation of a multimodal multitask vision foundation model for generalist ophthalmic artificial intelligence. *NEJM AI* 1, AIoa2300221 (2024). https://doi.org/10.1056/AIoa2300221

13. Aghabeigi Alooghareh, M. M., Sheikhey, M. M., Sahafi, A., Pirnejad, H. & Naemi, A. Deep learning for comprehensive analysis of retinal fundus images: Detection of systemic and ocular conditions. *Bioengineering* 12, 840 (2025). https://doi.org/10.3390/bioengineering12080840

14. Ju, L. et al. Synergic adversarial label learning for grading retinal diseases via knowledge distillation and multi-task learning. *IEEE Journal of Biomedical and Health Informatics* 25, 3709–3720 (2021). https://doi.org/10.1109/JBHI.2021.3052916

15. Pascal, L. et al. Multi-task deep learning for glaucoma detection from color fundus images. *Scientific Reports* 12, 12361 (2022). https://doi.org/10.1038/s41598-022-16262-8

16. Ting, D. S. W. et al. Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes. *JAMA* **318**, 2211–2223 (2017).

17. Abràmoff, M. D., Lavin, P. T., Birch, M., Shah, N. & Folk, J. C. Pivotal trial of an autonomous AI-based diagnostic system for detection of diabetic retinopathy in primary care offices. *npj Digital Medicine* **1**, 39 (2018).

18. Bellemo, V. et al. Artificial intelligence using deep learning to screen for referable and vision-threatening diabetic retinopathy in Africa: a clinical validation study. *The Lancet Digital Health* **1**, e35–e44 (2019).

19. Gargeya, R. & Leng, T. Automated identification of diabetic retinopathy using deep learning. *Ophthalmology* **124**, 962–969 (2017).

20. Ruamviboonsuk, P. et al. Deep learning versus human graders for classifying diabetic retinopathy severity in a nationwide screening program. *npj Digital Medicine* **2**, 25 (2019).

21. Gulshan, V. et al. Performance of a deep-learning algorithm vs manual grading for detecting diabetic retinopathy in India. *JAMA Ophthalmology* **137**, 987–993 (2019).

22. Poplin, R. et al. Prediction of cardiovascular risk factors from retinal fundus photographs via deep learning. *Nature Biomedical Engineering* **2**, 158–164 (2018).

23. Rim, T. H. et al. Prediction of systemic biomarkers from retinal photographs: development and validation of deep-learning algorithms. *The Lancet Digital Health* **2**, e526–e536 (2020).

24. Liew, G., Wang, J. J., Mitchell, P. & Wong, T. Y. Retinal vascular imaging: a new tool in microvascular disease research. *Circulation: Cardiovascular Imaging* **1**, 156–161 (2008).

25. Sabanayagam, C. et al. A deep learning algorithm to detect chronic kidney disease from retinal photographs in community-based populations. *The Lancet Digital Health* **2**, e295–e302 (2020).

26. Raghu, M., Zhang, C., Kleinberg, J. & Bengio, S. Transfusion: Understanding transfer learning for medical imaging. In *Advances in Neural Information Processing Systems* **32**, 3347–3357 (2019).

27. Yu, K. et al. UrFound: Towards universal retinal foundation models via knowledge-guided masked modeling. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2024*, 753–762 (2024).

28. Shi, D. et al. A multimodal visual–language foundation model for computational ophthalmology. *npj Digital Medicine* **8**, 381 (2025).

29. Silva-Rodríguez, J., Chakor, H., Dolz, J. & Ayed, I. B. A foundation language-image model of the retina: encoding expert knowledge in text supervision. *Medical Image Analysis* **99**, 103357 (2025).

30. Caruana, R. Multitask learning. *Machine Learning* **28**, 41–75 (1997).

31. Kendall, A., Gal, Y. & Cipolla, R. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 7482–7491 (2018).

32. Saito, T. & Rehmsmeier, M. The precision–recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE* **10**, e0118432 (2015).

33. Davis, J. & Goadrich, M. The relationship between precision–recall and ROC curves. In *Proceedings of the 23rd International Conference on Machine Learning*, 233–240 (2006).

34. Obuchowski, N. A. Nonparametric analysis of clustered ROC curve data. *Biometrics* **53**, 567–578 (1997).

35. DeLong, E. R., DeLong, D. M. & Clarke-Pearson, D. L. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics* **44**, 837–845 (1988).

36. Varoquaux, G. Cross-validation failure: small sample sizes lead to large error bars. *NeuroImage* **180**, 68–77 (2018).

37. Willemink, M. J. et al. Preparing medical imaging data for machine learning. *Radiology* **295**, 4–15 (2020).

38. Mongan, J., Moy, L. & Kahn, C. E. Jr. Checklist for Artificial Intelligence in Medical Imaging (CLAIM): a guide for authors and reviewers. *Radiology: Artificial Intelligence* **2**, e200029 (2020).